

International Trade Liberalization and Domestic Institutional Reform

Effects of WTO Accession on Chinese Internal Migration Policy

Yuan Tian, *Review of Economics and Statistics*, 2024

Central claim

Trade liberalization does not only reallocate production. It can also change the domestic institutions that constrain factor mobility.

Presentation design: economic mechanism → measurement → identification → results → interpretation

The paper in one slide

Research question

Did WTO-induced export-market liberalization make Chinese local governments relax Hukou-related migration restrictions?

Main finding

Prefectures facing larger export tariff reductions enacted more migrant-friendly regulations from 2001 to 2007.

Empirical setting

China after WTO accession: different prefectures were differentially exposed to foreign tariff cuts because their pre-WTO industrial structures differed.

Economic interpretation

Export opportunities raise the local return to labor mobility. Rigid internal migration policy becomes a binding, costly institution.

Key message

The paper treats migration regulation as an endogenous institutional response to trade-induced labor demand, rather than as a fixed friction.

Presentation roadmap

1. **Question and mechanism:** why trade liberalization should change local migration policy.
2. **Institutional setting:** China's Hukou system and decentralization around WTO accession.
3. **Data:** a new prefecture-level database of migration regulations.
4. **Identification:** shift-share exposure to foreign tariff cuts on Chinese exports.
5. **Evidence:** main results, heterogeneity, well-being, and migrant flows.
6. **Evaluation:** what is convincing, what remains hard to identify.

Why this is an economics question

Standard trade-liberalization margin

- ▶ price changes;
- ▶ output and export responses;
- ▶ firm entry, productivity, and reallocation;
- ▶ wage and employment effects.

This paper's margin

- ▶ domestic labor-market institutions;
- ▶ local policy responses;
- ▶ public-good access for migrants;
- ▶ endogenous reduction of mobility frictions.

Key message

The contribution is not another estimate of the China shock. It is direct evidence that trade exposure can induce observable institutional reform.

Institutional setting: Hukou as an internal mobility friction

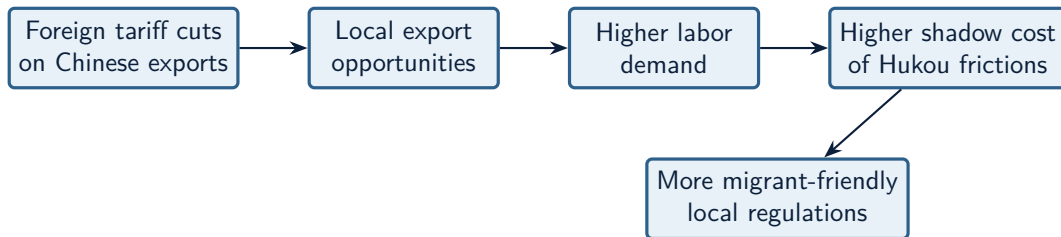
- ▶ Hukou assigns each citizen a location and an agricultural/nonagricultural status.
- ▶ Moving across prefectures requires temporary registration for legal migrant status.
- ▶ Even legal migrants face lower access to public goods: schools, medical insurance, social insurance.
- ▶ Before 2000, local governments had limited policy discretion.
- ▶ Around WTO accession, prefectures gained discretion to adjust migrant-related rules.

Why this setting is useful

- ▶ Same national trade shock.
- ▶ Heterogeneous local exposure.
- ▶ Decentralized policy outcome.
- ▶ Directly measurable regulations.

Source: Tian (2024), Sections II and III.

Mechanism: from export demand to institutional reform



Economic logic

Local governments internalize some gains from trade. When export liberalization raises the return to attracting workers, maintaining strict migration barriers becomes costly.

Measuring institutional reform: regulation texts

Corpus

- ▶ 4,273 migrant-related regulations, 1978-2016.
- ▶ 2,915 prefecture-level regulations.
- ▶ Keywords: migrant worker, non-Hukou population, temporary residence, Hukou.

Score

Each regulation is coded from -2 to $+2$; cumulative prefecture-level score measures migrant friendliness.

FIGURE 1.—THE MOST FREQUENT WORDS IN REGULATORY OR WORK-RELATED ISSUES AND LOCAL PUBLIC-GOODS-RELATED ISSUES

(a) Regulations on work-related issues



(b) Regulations on access to local public goods

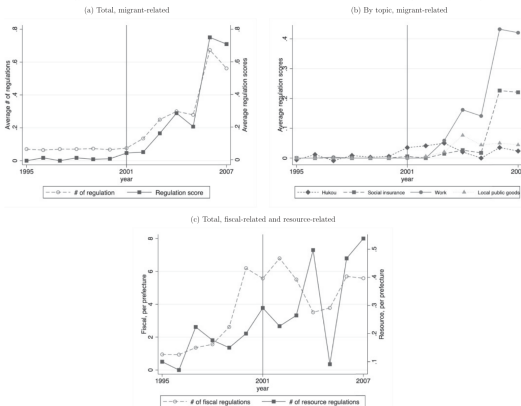


This figure shows the most frequent words of the migrant-related regulations on work topics (3,567 out of 4,273) and local public good access topics (286 out of 4,273). A larger font indicates a higher frequency. Appendix A.3 shows the corresponding word clouds for other topics: family planning policy, hukou, social insurance, and miscellaneous issues.

Source: Tian (2024), Section III and Figure 1.

What changed after WTO accession?

FIGURE 2.—THE NUMBER OF NEWLY-ENACTED MIGRANT-RELATED REGULATIONS AND REGULATION SCORES (1995–2007), TOTAL AND BY TOPIC (PANEL A AND PANEL B), AND THE NUMBER OF REGULATIONS IN THE FISCAL-RELATED AND RESOURCE-RELATED CATEGORIES (1995–2007, PANEL C)



Each dot is the average of 340 prefectures in a particular year. Panel a presents the trend in the prefecture-level average number of newly enacted migrant-related regulations (dashed line, vertical axis on the left)

Descriptive fact

Before 2001, prefecture-level migrant-related regulation activity is close to flat. After 2001, both the number and pro-migrant score rise sharply.

Where the increase comes from

The increase is mainly driven by work-related and social-insurance-related regulations.

Source: Tian (2024), Figure 2.

Validation: manual coding plus supervised NLP

- ▶ The author manually codes all 4,273 regulations.
- ▶ Two legally trained research assistants independently code a training set of 686 regulations.
- ▶ Three supervised methods validate the score: random forest, multilayer perceptron, and CNN with word embeddings.

Method	Validation	Corr. w/ manual
Random forest	OOB error 20.7%	0.734
MLP	val. acc. 83%	0.728
CNN	val. acc. 81%	0.712

Bottom line

The manually coded index is not mechanically reproduced, but the NLP scores are close enough to support the main empirical design.

Why this matters

The dependent variable is hand-built. Validation is essential because measurement error or author discretion could otherwise drive the result.

Source: Tian (2024), Section III; Online Appendix Table A3.

Trade shock: exposure to tariff cuts on exports

Prefecture-level tariff exposure

$$\Delta T_i^x = \sum_j s_{ij,2000} \Delta \tau_j^x$$

where $s_{ij,2000}$ is prefecture i 's initial employment share in industry j , and $\Delta \tau_j^x$ is the industry-level change in tariffs faced by Chinese exports.

Identification intuition

Same national WTO accession, different local exposure because prefectures had different pre-WTO industry mixes.

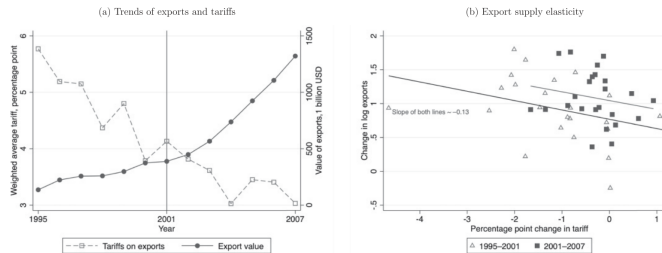
Sign convention

A larger tariff cut is a more negative ΔT_i^x . A negative coefficient means tariff cuts increase migrant friendliness.

Source: Tian (2024), Sections IV and VI.

Trade exposure: tariff cuts and export response

FIGURE 3.—THE TRENDS OF TARIFFS ON EXPORTS AND EXPORT VALUES (1995–2007) AND EXPORT SUPPLY ELASTICITIES (1995–2001 AND 2001–2007)



In Panel a, the dashed line presents trend of the weighted average of industrial-level tariffs on exports, where the weights are shares of exports in this industry. The industry-level tariff on exports is constructed as weighted average of destination-country tariffs on Chinese exports in the specific industry, where the weights are shares of exports in this destination country in 1995. The solid line presents trend of the value of exports. Panel b presents the relationship between percentage point change in tariff and change in log export for industries whose export values changed from 1995 to 2001 and from 2001 to 2007.

What the figure shows

- ▶ Export tariffs fell after WTO accession.
- ▶ Export values rose sharply.
- ▶ Industries with larger tariff declines show larger export growth.

Source: Tian (2024), Figure 3.

Main estimating equation

Long-difference specification

$$\Delta \sinh^{-1}(\text{RegScore}_i) = \alpha + \beta \Delta T_i^x + \Gamma' X_i + \varepsilon_i,$$

with the main period 2001-2007 and prefecture-level observations.

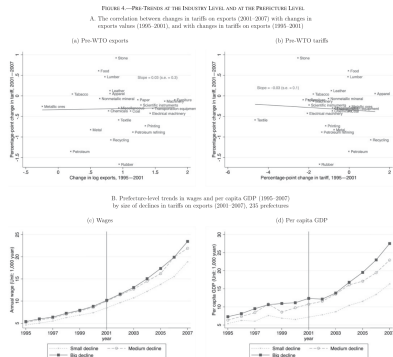
Controls and samples

- ▶ tariffs on imports and inputs;
- ▶ baseline regulation score;
- ▶ pre-WTO tariff changes, PNTR, MFA shocks;
- ▶ wage and GDP pre-trends in stricter specifications.

Threats to identification

- ▶ industries with larger tariff cuts may have different pre-trends;
- ▶ prefectures with larger exposure may have different pre-WTO growth;
- ▶ regulation changes may reflect broader local development policy.

Pre-trend diagnostics



Panel c and d present the relationship between tariff changes in the post-WTO period and pre-WTO tariff and export changes at the industry level. Each dot is a two-digit SIC industry. Panel (a) shows the relationship

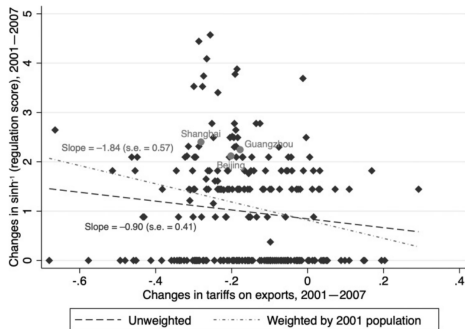
Interpretation

Post-WTO tariff declines are not predicted by pre-WTO export growth or earlier tariff declines; wage and GDP trends by exposure group are broadly similar before 2001.

Source: Tian (2024), Figure 4.

Main result: tariff cuts predict pro-migrant reform

FIGURE 5.—THE RELATIONSHIP BETWEEN PREFECTURE-LEVEL CHANGES IN TARIFFS ON EXPORTS AND REGULATION CHANGES (2001–2007), 248 PREFECTURES



Each dot is a prefecture. This figure presents the relationship between changes in tariffs on exports and changes in the inverse-hyperbolic-sine-transformed regulation score from 2001 to 2007. The dashed line is linear fitted line, unweighted, and the dotted line is the linear fitted line weighted by the 2001 population size. The sample includes 248 prefectures with GDP, wage, and population data from the 2001 *Prefecture Statistics Yearbook*.

Interpretation

Prefectures with larger export tariff reductions experienced larger increases in the IHS-transformed migration regulation score.

Magnitude

Upper-third tariff-cut prefectures increased by about 0.40 standard deviations more than lower-third prefectures.

Source: Tian (2024), Figure 5 and Table 1.

Main regression estimates

TABLE 1.—EFFECTS OF CHANGES IN TARIFFS ON EXPORTS ON CHANGES IN REGULATION SCORES (2001–2007), MANUAL CODING AND RANDOM FOREST CODING

Panel A.	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Dependent variable: $\Delta \sinh^{-1}$ (regulation score), 2001–2007, manual coding						
Δ tariffs on exports, 2001–2007	–1.04*** (0.30)	–0.91*** (0.26)	–1.48*** (0.33)	–1.20*** (0.32)	–1.60*** (0.50)	–1.38*** (0.49)	–1.02** (0.53)
Δ tariffs on imports, 2001–2007	–0.03 (0.04)	–0.01 (0.04)	–0.05 (0.03)	–0.04 (0.04)	–0.11** (0.04)	–0.08 (0.07)	–0.08 (0.07)
Δ tariffs on inputs, 2001–2007	–0.54*** (0.14)	–0.16 (0.20)	–0.53*** (0.14)	–0.21 (0.20)	–0.31** (0.14)	–0.04 (0.23)	–0.13 (0.23)
$\sinh^{-1}$ (regulation score), 2001	–0.09 (0.24)	–0.11 (0.22)	–0.10 (0.23)	–0.11 (0.22)	–0.12 (0.22)	–0.12 (0.21)	–0.15 (0.20)
PNTR shock, 2001–2007		4.67*** (1.44)		3.99*** (1.25)		2.89** (1.27)	1.03 (1.21)
MFA shock, 2001–2007		5.43*** (1.94)		4.89** (1.91)		4.20** (1.91)	4.86** (2.25)
Δ tariffs on exports, 1995–2001		–0.05 (0.16)		–0.04 (0.17)		–0.12 (0.21)	–0.17 (0.23)
Δ tariffs on imports, 1995–2001		–0.00 (0.02)		0.00 (0.02)		0.02 (0.03)	0.02 (0.02)
Δ tariffs on inputs, 1995–2001		–0.22** (0.10)		–0.20** (0.09)		–0.25** (0.12)	–0.14 (0.13)
Δ log wage, 1995–2001							0.69 (0.52)
Δ log GDP p.c., 1995–2001							0.65*** (0.22)
Observations	333	333	317	317	248	248	235
R-squared	0.09	0.14	0.11	0.15	0.08	0.11	0.15
Bootstrap p-value, Δ tariffs on exports	.006	.005	0	.003	.006	.014	.078
AKM1 p-value, Δ tariffs on exports	.004	0	.001	.002	.015	.022	.056
AKM0 p-value, Δ tariffs on exports	.024	.02	.02	.036	.036	.055	.051
Panel B.	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Dependent variable: $\Delta \sinh^{-1}$ (regulation score), 2001–2007, random forest coding						
Δ tariffs on exports, 2001–2007	–1.00*** (0.30)	–0.86*** (0.26)	–1.39*** (0.33)	–1.06*** (0.30)	–1.46*** (0.49)	–1.21** (0.47)	–0.89* (0.52)
Observations	333	333	317	317	248	248	235
R-squared	0.09	0.16	0.11	0.16	0.07	0.12	0.16

Baseline estimate

Column 1: export-tariff coefficient is about -1.04 and significant at the 1% level.

Robustness

The coefficient stays negative across controls, samples, and random-forest-based scores.

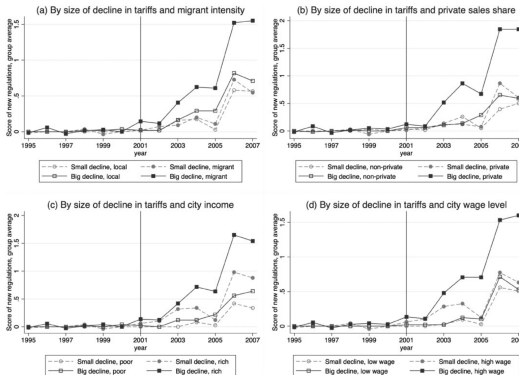
Sign

Tariff cuts are negative; $\hat{\beta} < 0$ means larger liberalization raises migrant friendliness.

Source: Tian (2024), Table 1.

Heterogeneity: when is the response strongest?

FIGURE 6.—AVERAGE PREFECTURE-LEVEL REGULATION CHANGES (1995–2007), BY THE SIZE OF THE CHANGE IN TARIFFS ON EXPORTS (2001–2007) AND BY MIGRANT INTENSITY (2001), 248 PREFECTURES



Panel a divides prefectures into four groups. The small-decline and local group represents prefectures whose post-WTO decline in tariffs on exports was below the median and migrant intensity was below the median.

Mechanism test

If reforms are driven by labor demand, responses should be stronger where additional migrant labor is more valuable.

Evidence

Large tariff-cut prefectures with higher migrant intensity, private-sector share, income, or wages show larger reform responses.

Source: Tian (2024), Figure 6 and Table 2.

Do these regulations matter de facto? Migrant well-being

TABLE 3.—THE RELATIONSHIP BETWEEN REGULATION SCORES AND MIGRANT OUTCOMES IN 2005

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A.	Dependent variable: Well-being of migrants, 2005						
	Unemploy. Insurance (rate) (1)	Pension (rate) (2)	Medical Insurance (rate) (3)	Terms of Contract (months) (4)	Monthly Income (yuan) (5)	School Enrollment (rate) (6)	Log # of Children (7)
$\sinh^{-1}$ (regulation score), 2005	0.01* (0.01)	0.02** (0.01)	0.02** (0.01)	0.09 (0.10)	29.27** (13.98)	0.00 (0.00)	-0.02 (0.01)
Well-being of local, 2005	0.57*** (0.09)	0.44*** (0.08)	0.40*** (0.05)	0.54*** (0.04)	0.37*** (0.11)	0.53*** (0.13)	0.10 (0.06)
Observations	246	246	246	246	246	246	246
R-squared	0.43	0.27	0.30	0.33	0.62	0.14	0.96
Panel B.	Dependent variable: Well-being of local population, 2005						
$\sinh^{-1}$ (regulation score), 2005	0.011 (0.011)	0.001 (0.008)	-0.001 (0.010)	0.386** (0.167)	16.553 (17.775)	0.005** (0.002)	-0.047* (0.027)
Observations	246	246	246	246	246	246	246
R-squared	0.43	0.45	0.43	0.22	0.57	0.095	0.90
Mean (s.d.) Y for migrant	.21 (.11)	.34 (.12)	.36 (.12)	4.57 (2.94)	923 (189)	.94 (.04)	4.48 (.99)
Mean (s.d.) Y for local	.37 (.16)	.60 (.14)	.59 (.15)	5.05 (2.88)	981 (336)	.95 (.02)	6.81 (.76)

Panel A regresses migrant well-being measures on the inverse-hyperbolic-sine-transformed regulation score and corresponding well-being measures of the local population, controlling for the log number of adult migrants and the log GDP p.c. in 2005. Panel B regresses local outcomes on the inverse-hyperbolic-sine-transformed regulation score, controlling for the log number of adult locals and the log GDP p.c. in 2005. All regressions are weighted by the 2005 prefecture population. The sample restriction is the same as in table 1 column 5, and two prefectures are not included since their GDP measures are absent in the 2005 *Prefecture Statistics Yearbook*. The mean (s.d.) of $\sinh^{-1}$ (regulation score), in 2005 is 0.41 (0.87). Standard errors are clustered at the province level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Outcome evidence

In 2005, migrants in more migrant-friendly prefectures have better outcomes, conditional on local residents' outcomes.

Outcomes

Social insurance, wages, contract length, contract status, and children's schooling.

Caution

Supporting evidence on implementation, not a clean causal estimate of enforcement.

Source: Tian (2024), Table 3.

Did trade and regulation changes affect migrant flows?

TABLE 4.—THE RELATIONSHIP BETWEEN CHANGES IN TARIFFS ON EXPORTS (2001–2007), REGULATION CHANGES (2001–2007) AND CHANGES IN MIGRANT FLOWS (2000–2010)

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Dependent variable:	Δ migrant share of population			Δ log # of migrants, short-dist.		
Δ tariffs on exports, 2001–2007	–7.11** (3.39)		–6.23* (3.22)	–1.34*** (0.31)		–1.18*** (0.27)
$\Delta \sinh^{-1}$ (regulation score), 2001–2007		1.03*** (0.29)	0.87*** (0.22)		0.18*** (0.04)	0.20*** (0.04)
Observations	248	248	248	238	238	238
R-squared	0.14	0.12	0.16	0.19	0.16	0.25
Mean (s.d.) of depent.		6.94 (5.77)			–0.84 (0.81)	
Panel B. Dependent variable:	Δ log # of migrants, medium-dist.			Δ log # of migrants, long-dist.		
Δ tariffs on exports, 2001–2007	–0.44** (0.20)		–0.39** (0.17)	0.09 (0.39)		0.16 (0.39)
$\Delta \sinh^{-1}$ (regulation score), 2001–2007		0.06* (0.03)	0.05* (0.03)		0.09** (0.04)	0.08* (0.04)
Observations	248	248	248	247	247	247
R-squared	0.59	0.57	0.59	0.12	0.08	0.13
Mean (s.d.) of depent.		1.60 (0.67)			0.85 (0.65)	

Reduced-form evidence

Tariff cuts and regulation changes are associated with larger migrant inflows from 2000 to 2010.

Important limitation

Trade affects migration both directly through wages and indirectly through policy. The paper is careful not to fully separate these two channels.

Source: Tian (2024), Table 4; Online Appendix C.11.

Robustness and alternative WTO-induced shocks

Measurement robustness

- ▶ manual score;
- ▶ random forest score;
- ▶ average NLP score;
- ▶ MLP and CNN scores;
- ▶ score levels and alternative regulation measures.

Trade-shock robustness

- ▶ controls for tariff changes on imports and inputs;
- ▶ PNTR uncertainty reduction;
- ▶ MFA quota removal;
- ▶ Autor et al.-style alternative trade exposure;
- ▶ industry-composition controls.

Key message

The result is not tied to one coding method or one narrow definition of WTO exposure.

How to read the paper's contribution

Contribution to trade and institutions

It provides direct evidence that trade liberalization can change domestic institutional rules, not merely outcomes under fixed institutions.

Contribution to migration and labor-market frictions

The paper treats Hukou restrictions as a policy margin that local governments adjust when the economic return to labor mobility rises.

Contribution to text-as-data measurement

The dependent variable is a large-scale hand-coded legal-regulation measure, validated with supervised NLP.

Critical evaluation

Strengths

- ▶ clear economic mechanism;
- ▶ new institutional dataset;
- ▶ subnational exposure to externally determined tariff cuts;
- ▶ strong validation of the policy measure;
- ▶ consistent heterogeneity evidence.

Limits

- ▶ de jure rules may not equal enforcement;
- ▶ high-growth prefectures may implement multiple reforms simultaneously;
- ▶ difficult to separate direct trade effects from policy-mediated migration effects;
- ▶ text score compresses heterogeneous legal content into one index.

Strongest causal claim: trade liberalization → observed pro-migrant regulation. Welfare and migrant-flow channels are persuasive but less cleanly separated.

Conclusion and discussion

Takeaway

WTO-induced export liberalization made internal labor mobility more valuable to Chinese prefectures, and prefectures responded by relaxing migrant-related regulations.

Broader lesson

Institutions that look politically sticky may adjust when the economic cost of maintaining them rises sharply.

Discussion questions

1. Is decentralised authoritarian governance central to the mechanism?
2. Are the reforms efficiency-improving labor liberalization, local fiscal competition, or both?
3. What evidence would distinguish regulation-driven migration from wage-driven migration?

Backup: heterogeneity regression table

TABLE 2.—HETEROGENEOUS EFFECTS OF CHANGES IN TARIFFS ON EXPORTS ON CHANGES IN REGULATIONS (2001–2007)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dependent variable: $\Delta \sinh^{-1}$ (regulation score), 2001–2007, manual coding							
Baseline characteristic:	Migrant intensity		Private firm share		Log(GDP p.c.)		Log(wage)	
Δ tariffs on exports 2001–2007	4.65 (3.65)	6.63 (3.90)	1.21 (0.72)	2.10** (0.91)	10.98* (5.97)	12.80** (5.86)	29.53 (18.72)	37.71** (18.26)
Baseline characteristic 2001	−0.87 (2.22)	−2.94 (2.49)	−0.50 (0.34)	−1.05** (0.44)	0.49*** (0.18)	0.48** (0.20)	0.54 (0.45)	0.46 (0.51)
Δ tariffs on exports $\times$ baseline chara.	−17.31 (10.93)	−21.89* (11.46)	−5.47*** (1.09)	−6.14*** (1.20)	−1.29* (0.68)	−1.47** (0.66)	−3.33 (2.07)	−4.18** (2.01)
Additional controls	X		X		X		X	
Observations	248	235	248	235	248	235	248	235
R-squared	0.09	0.16	0.13	0.20	0.23	0.27	0.15	0.21
Mean of baseline chara.	.34	.34	.55	.55	8.93	8.96	9.11	9.12
Sd. of baseline chara.	.05	.05	.23	.23	.64	.65	.28	.28

Columns 1, 3, 5, and 7 replicate table 1 column 5 and add a prefecture characteristic in 2001 and its interaction with the changes in tariffs on exports from 2001 to 2007. Columns 2, 4, 6, and 8 replicate table 1, column 7 and add a prefecture characteristic in 2001 and its interaction with the changes in tariffs on exports from 2001 to 2007. The baseline prefecture characteristic is the migrant intensity in 2001 in columns 1 and 3, and the baseline prefecture characteristic is the private firm share in 2001 in columns 2 and 4.

Source: Tian (2024), Table 2.

Backup: alternative NLP results in the appendix

- ▶ Random forest: interpretable decision-tree ensemble.
- ▶ MLP: neural-network classifier using words and word frequencies.
- ▶ CNN with word embeddings: uses word context as well as vocabulary.

	RF	MLP	
Training set	686	686	
Validation	OOB	holdout	h
Validation result	20.7% error	83% acc.	81
Corr. manual	0.734	0.728	

Source: Online Appendix Table A3.

Backup: interpreting the coefficient

Coefficient sign

ΔT_i^x is the tariff change; a larger liberalization is more negative. Therefore $\hat{\beta} < 0$ implies that larger tariff cuts raise migrant friendliness.

Baseline magnitude

Table 1 column 1 reports roughly $\hat{\beta} = -1.04$. A one-percentage-point larger tariff decline increases the IHS regulation-score change by about 1.04 units.

Distributional comparison

Big-decline prefectures have about a 0.36 p.p. tariff decline on average; small-decline prefectures about -0.04 p.p. The implied difference in regulation-score change is about 0.42 IHS units, or roughly 0.40 standard deviations.

Source: Tian (2024), Table 1 discussion.

Backup: migration-flow identification challenge

Why this is harder than the main result

$$\Delta \text{MigrantFlow}_i = \pi_0 + \pi_1 \Delta \sinh^{-1}(\text{RegScore}_i) + \Pi' X_i + u_i$$

- ▶ Regulations may attract migrants.
- ▶ Migrant inflows may also pressure governments to change regulations.
- ▶ Export tariff cuts affect migration directly through wages and indirectly through policy.
- ▶ Other local policies may move with migration regulations.

Key message

This is why the paper's migrant-flow section is best read as supporting mechanism evidence rather than the cleanest causal result.